From video observations to scene-conditioned human motion

Architecture draft / 25 Oct 2026 / unified EXO and full RGB-to-motion integration remain unvalidated

A Fixed upstream estimation

Existing components; freeze for downstream comparisons

RGB video

target identity
+ timestamps

Human observations

GroundingDINO / SAM2 / ViTPose / VIMO

MegaSam reconstruction

UniDepth + DepthAnything + RAFT / DROID

Alignment + gravity

MegaHunter + GeoCalib
shared metric frame

Observation package

camera poses + intrinsics
observed scene points
estimated body + validity

B Conditional motion diffusion

Existing EGO path + proposed unified observation adaptation

EGO: camera wearer

camera trajectory
+ allowed calibration

EXO: observed person

estimated 3D joints
+ confidence / missingness

Observation adapters

EGO existing
EXO adaptation proposed

Observed scene only

static prefix + current surfaces
remove target body

x_T ~ N(0, I)

Noisy motion

x_t + t

Denoiser: call 1

scene-free proposal
shared backbone theta

Body-relative query

predicted shape / FK
67 skin anchors
observed local geometry

Denoiser: call 2

same theta, x_t and t
scene-conditioned refinement
+ bounded correction

DDIM update

refined x_0 estimate
-> x_(t-1)

t > 0

t = 0

ONE DDIM CHAIN / 20 sampling steps / both denoiser calls receive the same x_t and t

Full-body motion

root trajectory + native decode

existing module path

proposed / unvalidated connection

EGO / EXO is defined relative to the target person. Ground truth enters supervision and evaluation only.

Offline setting; causal scene caching does not imply an online frontend.